Kuta Software - Infinite Precalculus

Name___________________________________

Geometric Sequences and Series

Date________________ Period____

Determine if the sequence is geometric. If it is, find the common ratio, the 8th term, and the explicit formula.

1) $-1,\ -3,\ -9,\ -27,\ ...$

2) $2,\ \frac{1}{2},\ \frac{1}{8},\ \frac{1}{32},\ ...$

3) $148,\ 1488,\ 14888,\ 148888,\ ...$

4) $0.75,\ 3,\ 12,\ 48,\ ...$

Given the explicit formula for a geometric sequence find the common ratio, the term named in the problem, and the recursive formula.

5) $a_n = -3 \cdot \left(\frac{1}{2}\right)^{n-1}$

Find a_{11}

6) $a_n = -1.5 \cdot (-2)^{n-1}$

Find a_{10}

Given two terms in a geometric sequence find the common ratio, the explicit formula, and the recursive formula.

7) $a_4 = -\frac{1}{4}$ and $a_1 = 2$

8) $a_5 = -24$ and $a_4 = -12$

Find the missing term or terms in each geometric sequence.

9) $...,\ 4,\ \_\_\_,\ \_\_\_,\ 108,\ ...$

10) $...,\ -25,\ \_\_\_,\ \_\_\_,\ \_\_\_,\ -\frac{1}{25},\ ...$